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PHYSICAL VERSUS CHEMICAL PROCESSES OF "WHITING" FORMATION IN THE BAHAMAS

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ABSTRACT: Whitings are common yet problematic occurrences of highly turbid water parcels among the relatively clear waters of shallow, tropical and subtropical carbonate platforms, banks and seas (e.g. Bahamas, Belize, Gulf of Carpentaria, Persian Gulf). In this paper, existing data on Bahamian whitings are scrutinized in an attempt to identify processes which may account for whiting phenomena worldwide.

An examination of the areal distribution of whitings on Great Bahama Bank shows that > 90% are associated with either mud or pellet-mud sediment facies. In view of this non-random distribution, several prominent hypotheses of whiting origin are considered: a) resuspension of fine-grained sediment by bottom-feeding fish (fish-muds); b) large-scale, instantaneous physicochemical precipitation events; c) large-scale carbonate precipitation incidental to planktonic algal blooms; d) bottom sediment resuspension as a result of turbulent tidal flow. Although none of the above hypotheses can be discounted completely, this paper proposes that the observed geographic and physical characteristics of whitings are consistent with predicted and observed characteristics of turbulent-flow systems; that whitings are the observable manifestation of the "bursting" cycle of turbulence production at flow boundaries.

The bursting process is a primary source of turbulence emanating from flow boundaries and can generate velocity excursions which deviate from mean flow velocity by a factor of 4. Published data for mean tidal current velocities on Great Bahama Bank range from 8 to 50 cm s^{-1}. Thus, maximum predictable velocities during the bursting process will range from 32 to 200 cm s^{-1}, quite in excess of empirically determined threshold sand-transport velocities (< 25 cm s^{-1}). Four-fold velocity excursions during turbulent flow increase shear stress up to 4-fold and generate 16-fold increases in lift force at the sediment water interface, enhancing bottom-sediment destabilization, particle entrainment and suspension. The visible, roiling nature of whiting water parcels is believed to be the expression of the above turbulent flow processes on Great Bahama Bank. Organization of turbulent boundary-layer flow into alternating low- and high-velocity "streaks" gives rise to alternating zones of active sediment-resuspension and deposition and may account for the development of subparallel digitations frequently observed within whitings.

An integrated, conceptual explanation of whiting distribution on Great Bahama Bank is proposed which considers the interaction of physical environmental energy and sediment textural characteristics across the bank-top. Despite high tidal-current, wind and wave energy flux (and sediment resuspension potential) near bank margins, whitings are rare because sediments are nearly devoid of mud-sized particles. Near Andros Island, carbonate-mud is plentiful but whitings are infrequent because deposition occurs in the tidal- and wind-energy shadow of the island. Only on the center of the bank-top is environmental energy (sediment resuspension potential) sufficiently high and carbonate-mud sufficiently abundant for whitings to originate. Evaluation of this proposed whiting generation concept requires an interdisciplinary understanding of chemical, physical, sedimentological and biological aspects of the Great Bahama Bank environment.

INTRODUCTION

The occurrence of isolated, elongate and turbid water parcels among the clear, shallow waters of tropical and subtropical carbonate platforms, banks and seas is a frequent event referred to as a "whiting". Whitings have been observed worldwide (Bahamas - Cloud, 1962a and b, Broecker & Takahashi, 1966, Morse *et al.* 1984, Shinn *et al.* 1989; Belize - Pusey, 1975, E.G. Purdy, personal communication, 1990; Gulf of Carpentaria - E.G. Purdy, per-

sonal communication, 1990; Persian Gulf - Wells & Illing, 1964). The highly localized nature of these whitings distinguishes them from basin-scale precipitation events observed in certain hypersaline lakes (e.g. the Dead Sea; Friedman, 1965) yet their origin remains enigmatic. Unfortunately, many whiting descriptions are anecdotal, containing little relevant information regarding seawater chemistry, physical oceanography, sedimentological or biological parameters. Thus, it is difficult to establish common features which might yield insight into the mechanism of whiting genesis.

In this paper, the available body of data on Bahamian whitings is reviewed. Fundamental characteristics of whitings are identified and an attempt is made to relate these features to common processes which may account for the occurrence of whiting phenomena worldwide.

Bahamian whitings contain as much as 21 mg l^{-1} of fine-grained (<8 um), suspended carbonate particles (dominantly aragonite) which have been variously interpreted as "fishmuds" (fine-grained sediments stirred into suspension by the activity of bottom-feeding fish; Ginsburg, 1956), products of active inorganic-carbonate precipitation within the whiting water parcel (Cloud 1962a and b; Shinn *et al.* 1989), incidental carbonate precipitation related to planktonic microbial blooms (Robbins & Blackwelder 1990 and 1992) or as fine-grained bottom-mud resuspended by purely physical processes (e.g. currents or waves; Broecker & Takahashi 1966; Morse *et al.* 1984). However, any hypothesis for the origin of whitings on Great Bahama Bank must identify processes which can account for 1) the observed temporal frequency of whitings and 2) the observed geographic distribution of whitings within bank-top waters.

The temporal frequency of whitings on Great Bahama Bank is not well known. Whitings have been observed in all seasons and in all weather conditions on the bank-top. Occasional storms (isolated high wind velocity events) are capable of generating platform-wide resuspension events causing the entire bank-top to be "whited" with fine-grained sediment (Neumann & Land 1975). However, whitings are known to originate and persist during extended periods of weak winds, thus excluding wind-induced turbulence as a solitary component in whiting origination.

Information regarding the geographic distribution of whitings on Great Bahama Bank was collected principally from satellite photographs obtained intermittently from 1983-1985 (personal collection of A.C. Neumann) and 1972-1975 (Morse *et al.* 1984). These data indicate that the distribution of whitings on Great Bahama Bank is not random. When superimposed on Purdy's (1963) map of Bahamian sediment facies (Fig. 1), > 90% occur over mud or pellet-mud sediment facies of central Great Bahama Bank (water depths over this region are generally <5 m). Indeed, of the 40 whitings observed by Shinn *et al.* (1989) over a three-year period, 39 (97.5%) occur over these facies. This is the same region where whitings were observed by Smith (1940), Cloud (1962a), Purdy (1963), Broecker & Takahashi (1966) and Morse *et al.* (1984). The persistence of this region as an area of whiting origination over many years is notable and comparable to the environmental setting of whiting occurrences in Belize where whitings are restricted to the shallow (<5 m deep), muddy (>60% material less than 0.125mm) northern shelf lagoon (Pusey, 1975; E.G. Purdy, personal communication, 1990).

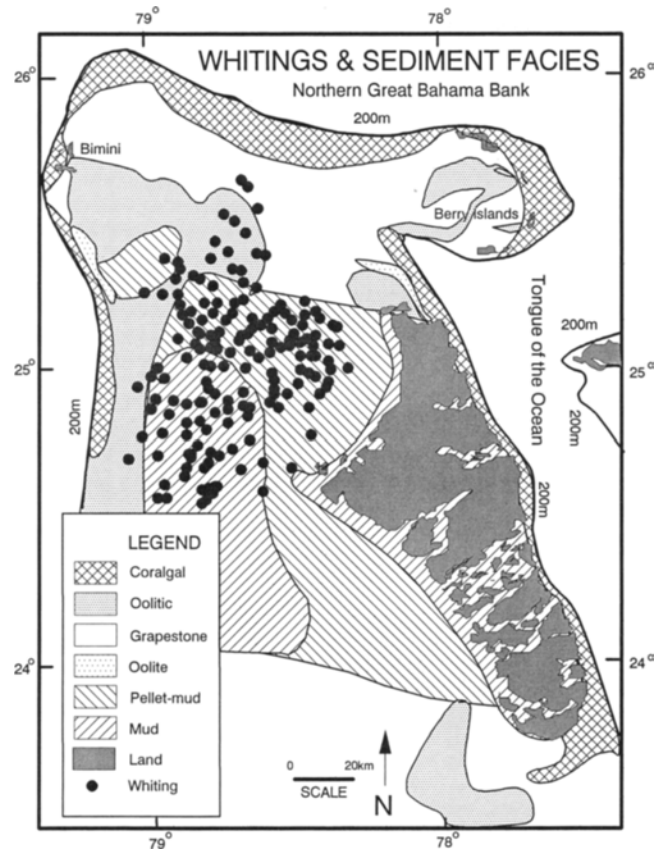


Fig. 1 Sediment facies of Great Bahama Bank (after Purdy, 1963) with whitings (closed circles) plotted from satellite photographs and bank-top observations (Shinn et al. 1989) superimposed. Notice that 90% of whitings occur over pellet-mud or mud sediment facies. The occurrence of a small number of whitings north of the major pellet-mud accumulation is coincident with Mackie Bank, an anomalously shallow, high-energy, linear sediment ridge of undetermined origin. The absence of whitings south of 24°-30' N latitude is due entirely to absence of data from this region, and should not be construed to indicate that whitings do not occur there.

In this paper, hydrographic, chemical and sedimentological factors associated with water parcels on Great Bahama Bank are reviewed in an attempt to provide insight into the whiting enigma. The four predominant hypotheses of whiting origin (fish-mud, inorganic chemical precipitation, biologically induced precipitation, turbulent resuspension) are then evaluated in view of these insights.

Textural Characteristics of Bahamian Carbonate Sediments

Great Bahama Bank
(From Purdy, 1963)

FACIES	GRAIN SIZE % > 0.125mm	% < 0.125mm
CORALGAL	89.2	10.8
OOLITIC	95.0	5.0
GRAPESTONE	95.5	4.5
OOLITE	98.1	1.9
PELLET-MUD	57.2	*42.8*
MUD	38.3	*61.7*

A theoretical argument is presented illustrating the probable effects of hydrodynamic processes (turbulent bursts) on bottom-sediment stability, entrainment and suspension during water movement over Great Bahama Bank. Further, it is suggested that these processes are sufficiently common (indeed, they are characteristic of turbulent flow systems) to account for both the observed spatial distribution and temporal frequency of whitings. However, evaluation of the whiting mechanism proposed herein requires an integrated, interdisciplinary view of various physical, chemical and biological components of the Great Bahama Bank environment.

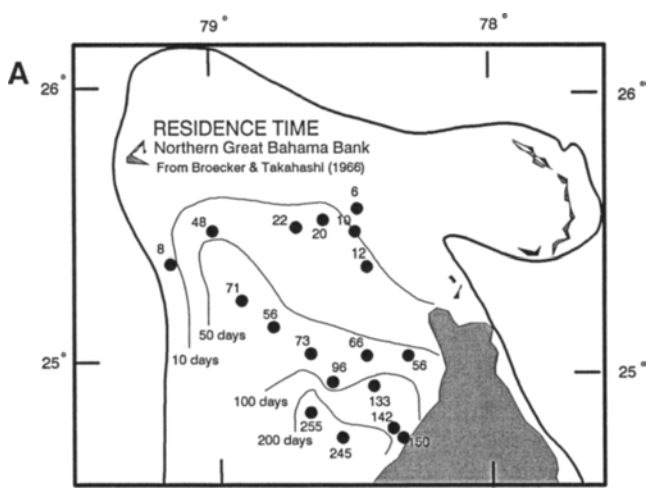
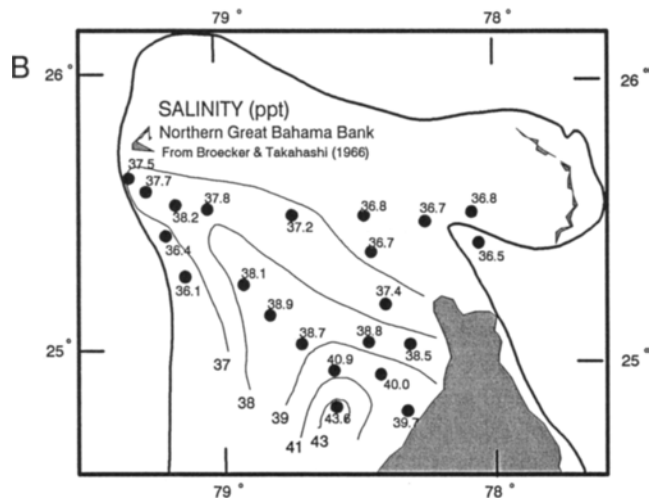


Fig. 2 A. Map showing residence time of waters on the northern Great Bahama Bank as determined from measurement of $\Delta^{14}C$ within bank-top water. Note the relatively rapid increase in water "age" leeward of Andros Island and compare with relatively youthful waters north of the Island (from Broecker & Takahashi 1966).

HYDROGRAPHIC SETTING OF WHITINGS

Smith (1940) was among the earliest to conduct comprehensive studies of the hydrography of Great Bahama Bank, relating physical/chemical properties of waters on the bank to prevailing surface winds, climatological processes and tidal flow across the bank-top. The development of a water parcel with elevated salinity (S = 43‰) on the interior of Great Bahama Bank leeward of Andros Island was documented. Development of this high-saline water (defined here as Great Bahama Bank Water, GBBW) was attributed to the effect of progressive evaporation as water moved onto the bank-top and into the lee of Andros Island. Although no attempt was made to estimate the residence time of water on the bank-top, it was recognized that development of 43‰ salinity required rather long residence.



B. Salinity contours (‰) of northern Great Bahama Bank waters. The progressive southward increase in salinity results from prolonged evaporation as water transits the bank-top and becomes "ponded" in the lee of Andros Island (from Broecker & Takahashi 1966).

Broecker & Takahashi (1966) and Morse *et al.* (1984) identified important chemical parameters that aided in evaluating hydrographic characteristics of waters on Great Bahama Bank. Measured ^{14}C activity of bank-top waters confirmed the suggestion of Smith (1940) that water age increased progressively from the bank margins toward the bank interior, with a maximum residence time of approximately 250 days leeward of Andros Island; waters north of Andros Island typically show residence times of <20days (Fig. 2a). Salinity measurements (Fig. 2b) were similar to those of Smith (1940), showing a progressive increase with water age on the bank-top. Thus, the most saline waters were associated with the oldest water parcel leeward of Andros Island, a trend confirming the effect of prolonged evaporation as water moved toward the bank interior from the open-ocean margins. Accurate definition of GBBW residence

time is important in evaluating chemical trends within bank-top waters below.

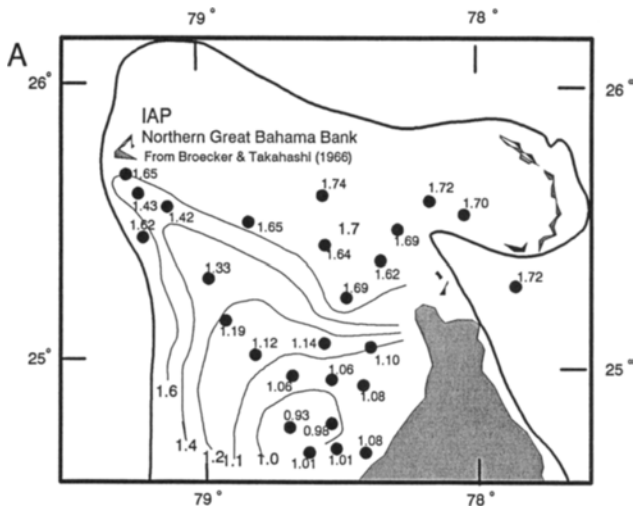


Fig. 3 A. Ion activity product (x 10^{-8}) of calcium carbonate from Broecker & Takahashi (1966) on the Great Bahama Bank. Ω *- aragonite is calculated from these data as IAP/K_{sp}, assuming K_{sp} = $10^{-8.33}$. It is important to note that* Ω *declines progressively from 3.5 to 1.95 from the bank margins to the central bank-top west of Andros Island.*

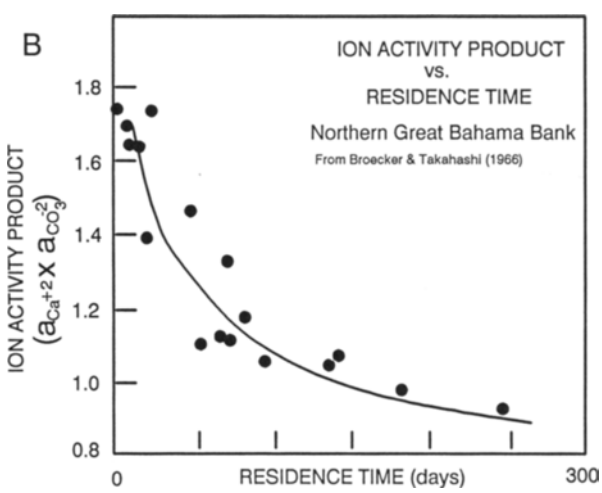


Fig. 3 B. Ion activity product (IAP x 10^{-8}) vs. residence time on the Great Bahama Bank (Broecker & Takahashi 1966). Approach of the carbonate system to thermodynamic equilibrium (Ksp = $10^{-8.33}$) occurs over the 250 day residence of the waters on the bank-top, as defined by the second-order rate function of Broecker & Takahashi (1966). This result suggests that whiting-scale, instantaneous precipitation events are kinetically unlikely.

FISH-MUDS AND WHITINGS

It has long been suggested that whitings are generated by the activity of bottom-feeding fish (e.g. mullet, blue runner or bonefish). Indeed, this idea is deeply entrenched in the common wisdom of the Bahamian people. According to this idea, schools of bottom-feeding fish ingest large quantities of bottom-sediment, extract the edible components and eject remaining fine-grained sediment into the water column through their gills or use their heads to churn bottom sediments (an activity referred to as "rooting") to generate whitings. Fish such as mullet and bonefish are known to feed in such a manner and the fish-mud hypothesis cannot be disregarded. However, the fish-mud hypothesis suffers from several unresolved issues. First, no credible witness has been able to produce tangible documentation of fish engaged in the act of generating whitings, despite rather sophisticated attempts to do so (Shinn *et al.* 1989). Second, there are presently no data to indicate that fish occur in appropriate numbers on Great Bahama Bank to account for the large number of whitings which can be observed at any one time. Third, the largest whitings (up to several kilometers in length and breadth) would seem to require large schools of fish which have never been reported from the shallow platforms of the Bahamas.

PHYSICAL CHEMISTRY OF WHITINGS

A re-examination of the physical chemistry of waters on Great Bahama Bank (Broecker & Takahashi 1966; Morse *et al.* 1984) with respect to the relatively long residence times provides a frame of reference from which the chemical basis for whiting phenomena may be evaluated. A review of relevant chemical parameters follows.

RELATIVE ARAGONITE SATURATION

Relative aragonite saturation Ω; calculated from ion activity products in Fig. 3a) can be determined from the data of Cloud (1962b) and Broecker & Takahashi (1966). Both of these studies showed that open-ocean surface-waters of the Bahamas region are approximately 3.5 times saturated with respect to aragonite (i.e. Ω = 3.5) and that Ω progressively declined as waters transited the bank-top. In fact, the lowest measured saturation (Ω = 1.95; IAP = 0.93 x 10^{-8}) occurred west of Andros Island and was associated with the oldest water parcel (approximately 250 days; Takahashi 1966; Morse *et al.* 1984; Broecker & Fig. 3b). Thus, GBBW appeared to obey fundamental equilibrium thermodynamic principles (i.e. supersaturation was reduced through $CaCO_3$ removal) in its journey across the bank-top. It is important to note, however, that Cloud (1962b) interpreted the transition from 250% supersaturation at the bank margins to 95% supersaturation at the bank-interior to be the result of numerous instantaneous precipitation events (i.e. whitings), whereas Broecker & Takahashi (1966) provided data suggesting a link between kinetic parameters and equilibrium chemistry of GBBW; that is, that the approach of the carbonate system to thermodynamic equilibrium was a rate-limited process requiring >250 days (Fig. 3b). In fact, Broecker & Takahashi (1966) were able to derive a relatively simple second-order rate function indicating that the rate of $CaCO_3$ removal was proportional to the degree of supersaturation (i.e. faster $CaCO_3$ removal in more supersaturated solutions). Whereas the precise mechanism

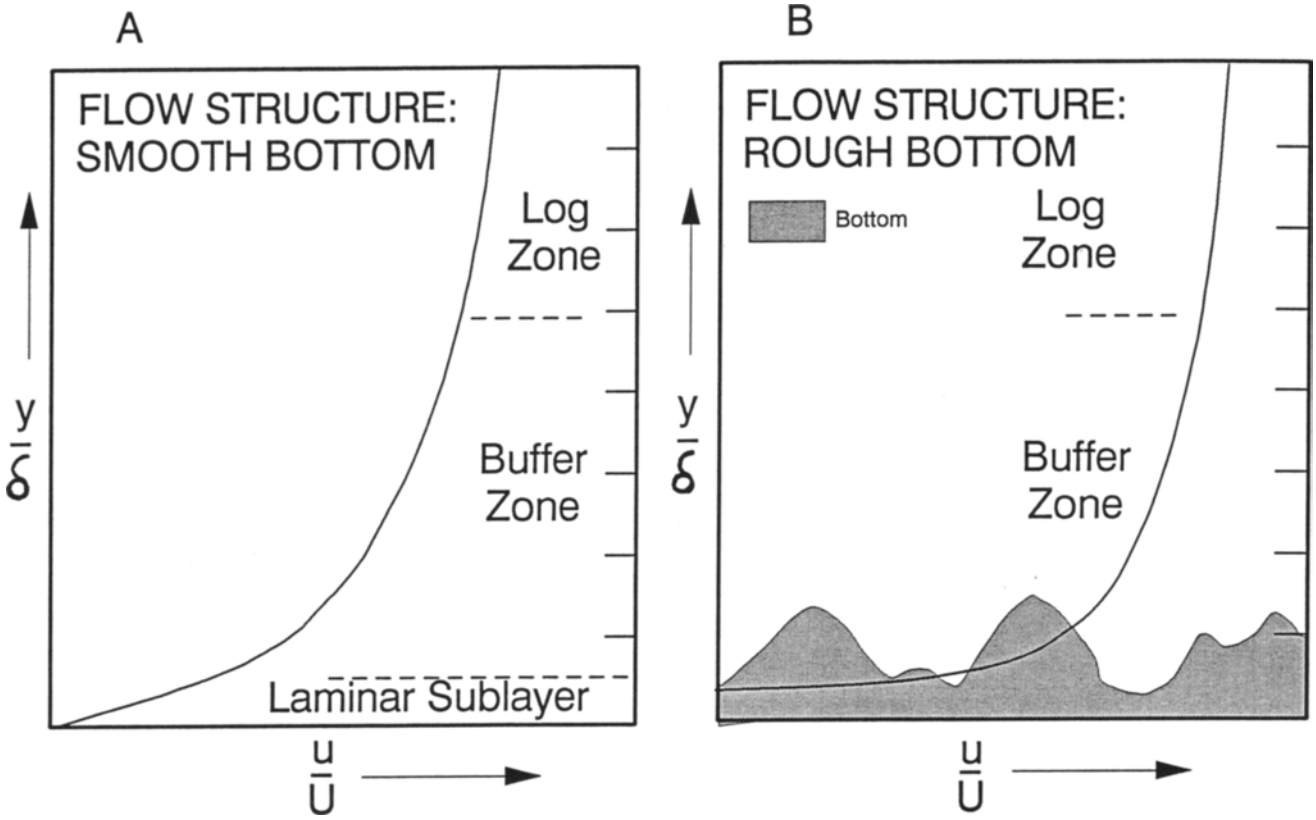


Figure 4. Schematic diagrams of the structure of the boundary layer over smooth (A) and rough (B) walls and showing velocity profile through the boundary region. Horizontal axis shows instantaneous velocity (u) as a proportion of mean velocity of flow (U). Vertical axis represents relative position within boundary-layer as the proportion of height above wall (y) vs. boundary layer thickness (δ). Note that representation of the dependent and independent variable in this way results in dimensionless values, implying that absolute scaling phenomena are essentially irrelevant in defining the structure of the boundary layer. Also, note that the laminar or viscous sublayer may be missing altogether during flow over rough walls (B). Not shown on this figure is the wake region, which occupies up to 80% of the boundary layer thickness (from Daily & Harleman 1966).

(whether physico-chemical or biologically controlled) responsible for the close correspondence of data to this kinetic function cannot be identified from the data alone, the close agreement of data to second-order kinetics suggests purely physico-chemical (equilibrium thermodynamic) control of carbonate extraction from GBBW. In essence, there is no reason to suppose that biologically controlled removal rates would adhere to second-order kinetics; it is more likely that such removal would depend significantly on the abundance of various carbonate-secreting organisms and their temporal or spatial distributions and interactions. Further, the close correspondence of chemical data to the described second-order kinetic function is an indication that chemical gradients within GBBW are thermodynamically and kinetically conducive to long-term inorganic precipitation of $CaCO_3$ but not to instantaneous, whiting-scale precipitation phenomena.

ALKALINITY OF GREAT BAHAMA BANK WATERS

Measurement of changes in alkalinity ($[HCO_3^-] + 2[CO_3^{-2}]$) between whiting water parcels and adjacent clear waters have been proposed as a means of evaluating whether or not active precipitation is occurring within the whiting water parcel. Morse et al. (1984) attempted to measure these changes, but were unable to find any systematic, significant variations which would indicate instantaneous, large-scale precipitation. Instead, measured variations in alkalinity of bank-top waters were found to conform very well to $CaCO_3$ removal rates and salinity changes associated with water-mass age. Essentially, the major alkalinity variations within GBBW are related to progressive removal of $CaCO_3$ over time, and the observed alkalinity variations obey the same kinetic relations that were identified by Broecker & Takahashi (1966). Namely, that $CaCO_3$ precipitation (and hence, alkalinity change) is proportional to the degree of supersaturation of GBBW. Thus, alkalinity data of Morse *et al.* (1984) provide additional evidence that GBBW is not conducive to whiting-scale precipitation events.

CO_2 PARTIAL PRESSURE

The partial pressure of CO_2 in bank-top waters was presumed to be a sensitive indicator of potential precipitation by Broecker & Takahashi (1966) and by Morse et al.

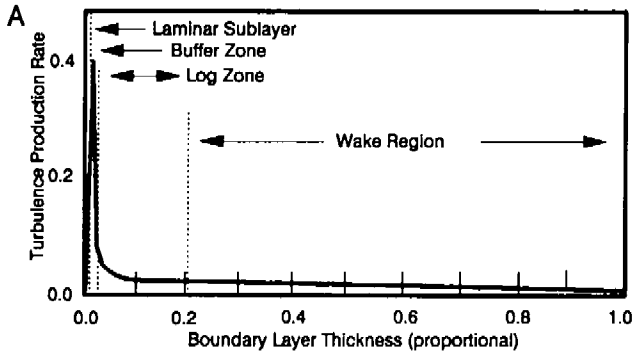


Figure 5A. Turbulent energy production rate per unit volume in a typical boundary layer. Note the dramatic peak in turbulent production associated with the innermost region of the boundary-layer (buffer layer) and the virtual absence of turbulence energy contribution from other regions (from Kline et al. 1967).

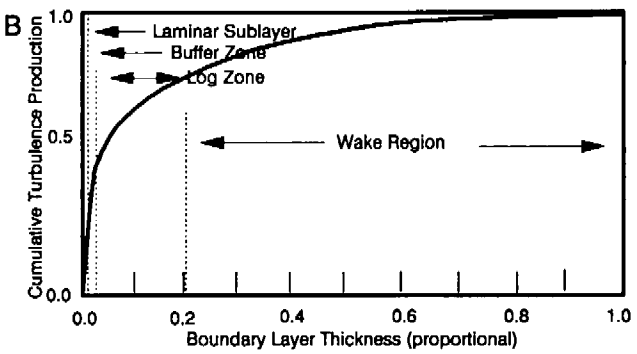


Figure B. Cumulative turbulent energy production rate in a typical boundary layer. This plot shows clearly that ca. 50% of turbulent energy production in the boundary layer is contributed by dynamic processes within the relatively thin buffer region. Note again that the large wake region contributes relatively little turbulent energy to the system (from Kline et al. 1967).

(1984). In particular, Broecker & Takahashi (1966) estimated that if all aragonite needles within whitings were forming through inorganic precipitation, the measured change in PCO_2 between whiting water and adjacent clear water would be 67 ppmv (parts per million by volume). In fact, their maximum measured PCO_2 within whitings was only 25% (15 ppmv) of this value and indicated that perhaps only 25% of the aragonite contained in a whiting was precipitated within the whiting water parcel. This value agrees well with the 15% primary precipitate estimated using ^{14}C activity by Broecker & Takahashi (1966) and 25% determined from stable isotopic data ($\delta^{13}C$ and $\delta^{18}O$) by Shinn *et al.* (1989). These estimates may be taken to suggest that whitings are, in fact, combinations of resuspended bottom sediment and precipitate; supposedly, suspension of fine-grained bottom sediments would provide suitable nuclei for enhanced active precipitation of aragonite from the water column. However, such precipitation must be viewed as incidental (secondary) to the process of sediment resuspension.

^{14}C ACTIVITY OF WHITING ARAGONITE

Measured ^{14}C activity of whiting aragonite has consistently yielded ages of 200-300 years (similar to ages determined for bottom muds; Broecker & Takahashi 1966; Morse *et al.* 1984; Shinn *et al.* 1989). Some authors view such determinations as inexplicable anomalies (Robbins & Blackwelder 1992). However, a review of ^{14}C systematics in marine environments shows that whiting radiocarbon assays are reasonably explained using available geochemical data; an observation which places important constraints on possible sources of fine-grained carbonate within whitings.

It is well-established that marine (carbonates both biogenic and inorganic) are enriched in ^{14}C relative to the waters from which they are precipitated as a result of kinetic isotopic fractionation (Craig, 1954; Libby, 1955). Since this enrichment always results in anomalously "young" apparent ages for carbonates, it must be concluded that "old" age determinations for whiting sediment reflect a significant fraction of genuinely "old" carbonate debris. Indeed, since the studies discussed previously (Broecker & Takahashi, 1966; Morse *et al.* 1984; Shinn *et al.* 1989) have shown that (at best) only 25% of material suspended within whitings can be attributed to active inorganic precipitation within the whiting water parcel, the relative ^{14}C-ages of "new" and "old" carbonate contributing to bulk age determinations of whiting sediment may be estimated using a simple end-member mixing equation of the form:

$$0.25(T_n) + 0.75(T_o) = -200 \quad (1)$$

where T_n is the bulk ^{14}C-age of "new" whiting carbonate, T_o is the bulk ^{14}C-age of "old" whiting carbonate and -200 is the measured bulk ^{14}C-age of all whiting sediment (negative sign used to indicate years before present). Beginning with the assumption that the 25% whiting sediment attributed to active inorganic precipitation yields a ^{14}C assay with zero age (i.e. $T_n = 0$), the equation solves directly for T_o :

$$T_o = -200/0.75 = -266.7$$

Thus, the bulk ^{14}C-age of the 75% of whiting material not directly attributable to active precipitation must be at least 267 years before present.

Open-air testing of atomic weapons during the 1950's and 1960's doubled the atmospheric inventory of ^{14}C (Damon *et al.* 1978). Some of this radiocarbon has been incorporated into ocean surface waters with the result that "modern" carbonates (living calcareous organisms and precipitates) have become enriched in ^{14}C relative to NBS oxalic acid ^{14}C standards to the extent that they yield "future" dates (i.e. radiocarbon assays of marine carbonate material precipitated since the 1960's are converted to non-zero, positive ages). While the magnitude of this effect is variable, it is omnipresent in shallow marine environments (such

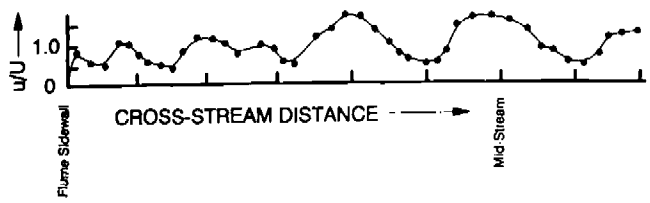


Figure 6. Instantaneous transverse (i.e. across stream) velocity profile in experimental boundary layer flow. Vertical axis is relative velocity expressed as the ratio of boundary layer velocity (u) to total mean velocity (U). Horizontal axis is span-wise distance. Notice the organization of transverse flow into alternating low- and high-velocity "streaks" which are subject to broad temporal oscillations. Organization of flow into streaks is the first stage of the bursting cycle (from Kline et al. 1967).

as the Bahamas) and always in the positive direction. Taking this factor into account in (1), T_o becomes positive, with the result that T_o must become even older than 267 years in order to satisfy the empirical requirement that bulk whiting sediment yield a ^{14}C-age of -200 years. This simple calculation reinforces the argument that "old" ^{14}C-ages derived for whiting carbonate are genuinely old and not the result of unknown or enigmatic fractionation mechanisms.

SUMMARY OF CHEMICAL DATA

The critical contribution to the understanding of the carbonate system on the Great Bahama Bank was the determination of the relatively long residence time of bank-top waters by Broecker & Takahashi (1966). This parameter was underestimated by Cloud (1962a and b) and by Traganza (1967) in their analyses of calcium-carbonate precipitation west of Andros Island, and it seems to have been ignored by Shinn *et al.* (1989) and by Robbins & Blackwelder (1990 and 1992). However, knowledge of this parameter permitted the details of kinetic factors controlling carbonate precipitation to be established. Of particular importance was the observation that Ω -aragonite declined progressively from the bank margins to the bank interior, coincident with the increase in relative water age. That is to say that Ω declined from approximately 3.5 (250% supersaturation) near the bank margins to a value of 1.95 (95% supersaturation) in the oldest water parcel west of Andros Island. The value of Ω= 1. 95 was particularly interesting because it demonstrated that GBBW approached equilibrium $CaCO_3$ concentration during its 8 month (250 day) residence on the bank-top. This feature is somewhat remarkable given the rather high probability for non-steady state perturbations (e.g. major storms which can replenish bank-top waters) during any 8 month period. In addition, it was demonstrated that the rate of $CaCO_3$ removal was proportional to the degree of supersaturation of the solution. The observed geographic distribution of whitings (Fig. 1) shows that whitings occur most frequently beneath the least saturated waters (Fig 3a). Thus it seems highly unlikely that instantaneous precipitation of the scale necessary to sustain whitings could occur.

The pH and alkalinity of waters surrounding and within whitings is known to be nearly constant (Broecker & Takahashi 1966; Morse *et al.* 1984). This observation contradicts theoretical calculations showing that inorganic precipitation of aragonite within the whiting will produce significant changes in both pH and alkalinity of the whiting water parcel (Morse *et al.* 1984). The near-constancy of these measurements indicates that chemical conditions in waters around and within whitings are not particularly well-suited to whiting-scale inorganic precipitation and that whiting phenomena must result from other than strictly chemical processes. Finally, consistent $\Delta^{14}C$ measurements indicate that whiting sediment is genuinely older than 200 yr. This result implies that much whiting carbonate is derived from pre-existing sources, most likely bottom sediments (Broecker & Takahashi 1966).

BIOLOGICALLY INDUCED PRECIPITATION

Recent observations of fine-grained calcium carbonate associated with organic floccules (tentatively identified as planktonic microbes) filtered from whitings have led to the suggestion that whiting phenomena result from large-scale carbonate precipitation following microbial blooms (Robbins & Blackwelder 1990 and 1992). The documentation of fine-grained carbonate with organic debris is interesting as it relates to the problem of lime-mud genesis, but its utility as an explanation of the whiting phenomenon seems limited. Significantly, no data are currently available to indicate that isolated, local blooms of sufficient magnitude occur with the required frequency to account for the observed distribution of whitings. In fact, Smith (1940) and Cloud (1962a) showed that bank-top waters are nearly devoid of essential nutrients, especially phosphate (usually <lumol l^{-1}; Cloud 1962a) and it is difficult to imagine sustained, largescale productivity in such nutrient-deficient waters.

Depletion of PCO_2 as a result of rapid photosynthetic uptake within blooms would be expected and such changes would promote calcium-carbonate precipitation. However, dramatic CO_2-depletions have not been observed within whitings. Indeed, PCO_2 measurements by Morse *et al.* (1984) indicate that for bank-top waters, PCO_2 is generally 1.5 times higher than in open-ocean waters, agreeing with estimates of Traganza (1967) that CO_2 production through organic-matter respiration and carbonate precipitation in GBBW outpaces CO_2 removal through photosynthesis and air-sea gas exchange.

Finally, while "ultrastructural" observations of fine-grained carbonate associated with organic debris (Robbin

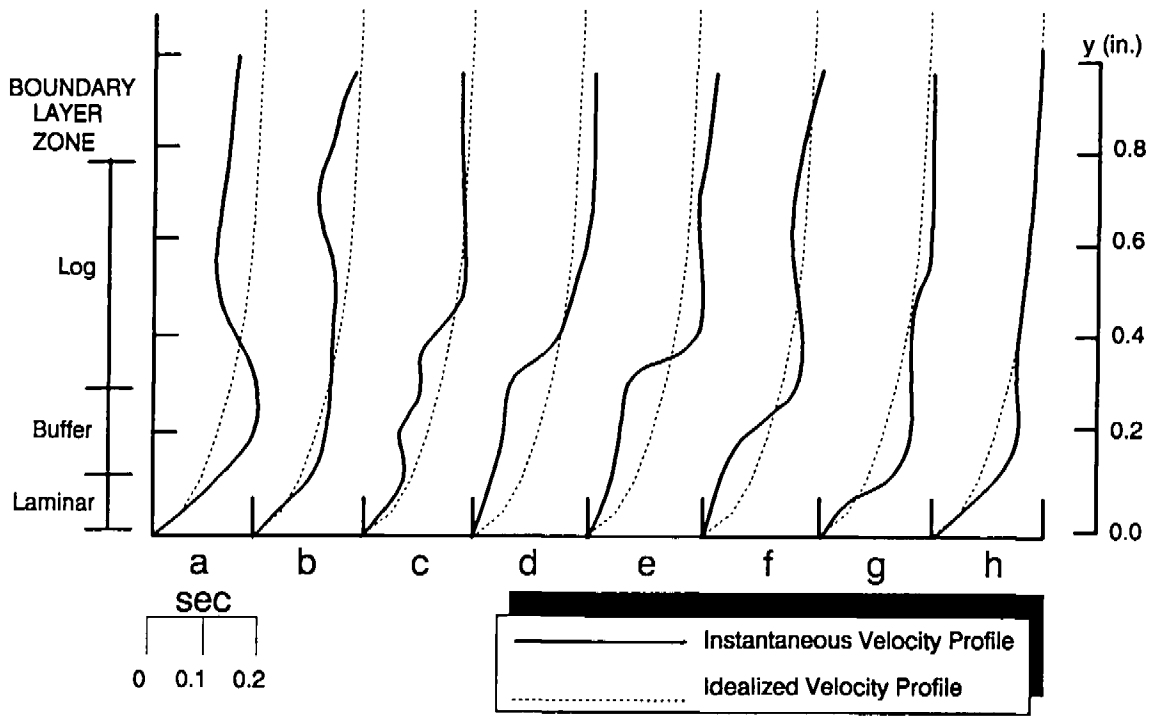


Figure 7. Time-series of instantaneous velocity profiles through the boundary layer. Examination of these plots from right to left shows the initiation of low-velocity streak lifting (ejection phase of bursting) visible as a velocity inflection propagating upward from the wall region (g through b). The lifting (ejection) phase is followed abruptly by in-rush of high-velocity fluid (sweep phase of bursting) visible as a prominent bulge in the last velocity profile (a). It is during this phase of bursting that instantaneous velocity excursions as much as 4 times mean flow may occur (see Fig. 8). Following this phase of the bursting event, the instantaneous velocity profile acquires a shape similar to the mean velocity profile until the next bursting event begins (from Kim et al. 1971).

& Blackwelder 1990 and 1992) provide insight into one possible mode of carbonate-mud production, such studies provide little insight into the "megastructural" phenomenon which is a whiting. Thus, the proposed biologically-induced mechanism of whiting formation fails to explain much that is known concerning the geographic and temporal distribution of whitings as well as physical characteristics of individual whitings.

WHITINGS AS TURBULENT-FLOW PHENOMENA

Among the remaining hypotheses which have been proposed to explain the occurrence of whitings on Great Bahama Bank, the suggestion that whitings result from physical resuspension of fine-grained bottom-muds has never been adequately tested. As a result of crude measures of tidal-current flows (Shinn *et al.* 1989), there has developed a general consensus that water movement on the bank-top is too sluggish to provide the required energy for sediment resuspension. In the following discussion, some elementary principles of the structure and dynamics of turbulent flow are outlined, and the relation of these principles to sediment entrainment and suspension are presented. It is intended that this discussion will provide the theoretical foundation needed to consider the role of turbulent-flow phenomena in the entrainment and continued suspension of fine-grained sediment to yield whitings. In addition, it is important to note that much knowledge of turbulent flow systems is derived from experimental studies using flumes. As such, the scale of observed turbulent flow phenomena is likely much smaller than that to be expected in natural systems. However, fluid dynamicists are well aware that processes of fluid flow display a high degree of selfsimilarity. In fact, the general tradition of representing data in fluid-dynamic studies as dimensionless values implies that absolute-scaling phenomena are largely irrelevant in the application of theoretical constructs to natural systems (Middleton & Southard 1984). On this basis, it is suggested that the physical characteristics of whitings are consistent with predicted and observed features of turbulent-flow systems on relatively large scales, and that these turbulent-flow processes are sufficiently common to account for the geographic and temporal distribution of whitings.

STRUCTURE OF TURBULENT BOUNDARY LAYERS

Numerous investigations of the structure of turbulent flow have revealed that turbulence production is intimately related to flow characteristics near fluid and solid boundaries (Daily & Harleman 1966). Indeed, the struc-

ture of the boundary layer has been described in some detail as a result of flow visualization experiments (Kline *et al.* 1967; Kim *et al.* 1971; Grass 1971). These experiments show that boundary layers can be subdivided into several regions which display characteristic flow behavior (Fig. 4; Daily & Harleman 1966). For a smooth boundary, there exists a narrow region where viscous forces dominate and flow is laminar (viscous or laminar sublayer, Fig. 4a). For water (relatively low viscosity), this region is generally exceedingly thin (Kline *et al.* 1967) over a smooth boundary and may be missing altogether over rough boundaries (Fig. 4b; Daily & Harleman 1966). Adjacent to the viscous sublayer there is a zone in which laminar flow decays rapidly into intense turbulence (the so-called buffer zone, Fig. 5a) which appears to contribute up to 50% of the cumulative turbulence production in the system (Fig. 5b; Kline *et al.* 1967). Finally, there are two broad regions (logarithmic zone and wake zone) over which mean flow velocity increases geometrically to its maximum magnitude. It is interesting to note that the wake region occupies up to 80% of the boundary layer, yet contributes <20% of the total turbulence production (Fig. 5b; Kline *et al.* 1967). Thus, concentration of current measurements within this region are unlikely to reveal significant flow anomalies associated with turbulence production. Indeed, flow visualization experiments (Kline *et al.* 1967; Kim *et al.* 1971) confirm the dominance of the near-wall region (buffer zone) in turbulence production and have permitted additional detailed descriptions of the process of turbulence production by intermittent low-velocity ejection/high-velocity sweeps within the fluid ("bursting").

DYNAMICS OF THE "BURSTING" CYCLE

Flow visualization studies by Kim *et al.* (1971) examined the detailed sequence and structure of the burst cycle in turbulent flow. Their observations indicated that virtually all turbulence production within flows occurred during the burst cycle at the boundary layer and established the importance of understanding the dynamics of bursting.

The burst cycle appears to begin with the organization of boundary-layer flow into alternating low and high velocity "streaks", which can be observed in transverse across stream velocity profiles (Fig. 6; Kline *et al.* 1967). As this flow pattern becomes organized, low-velocity streaks are seen to be lifted from the wall region. Lifting of low-velocity streaks into the stream flow can be observed on time series of nstantaneous velocity profiles (Fig. 7) as a velocity defect (inflection point) rising from the wall region. Following ejection of the low-velocity streak from the boundary region, the instantaneous velocity profile (Fig. 7) shows an in-rush (sweep) of higher velocity fluid, presumably derived from transverse high-velocity streaks adjacent to the lifted low velocity streak. This in-rush of higher velocity fluid stabilizes the instantaneous velocity profile and the lowvelocity ejection/high-velocity sweep (bursting) cycle is completed.

Bursting phenomena can be identified as intermittent, large-amplitude oscillations in time-series plots of Instantaneous boundary-layer flow velocity (Fig. 8). It is particularly important to note the magnitude of velocity fluctuations within the burst cycle, as these fluctuations will enhance entrainment and suspension of sediments from the boundary by increasing bottom shear stress and lift force (Middleton & Southard 1984). Grass (1971) reported velocity excursions as large as 3 to 4 times the mean flow velocity, even in relatively slow-moving fluids. For example, in a mean current of 15 cm s^{-1} it would be common to experience transient excursions as large as 45-60 cm^{-1}. Bottom shear stress is proportional to the velocity gradient in the nearwall region of flow (Middleton & Southard 1984) and assuming this gradient is linear over the narrow zone of burst activity, an instantaneous 4-fold increase in flow velocity will result in a 4-fold increase in bottom shear stress. However, flow over rough bottoms will create lift forces which are related by:

$$\Delta P = C_L(\sigma/2)u^2 \qquad (2)$$

where Δ is the vertical pressure change, C_L is the lift coefficient (constant), σ is fluid density and u is instantaneous velocity (Einstein & El-Samni 1949). Thus, 3-to 4-fold increases in instantaneous velocity associated with bursting will result in 9- to 16-fold increases in the lift force affecting the wall region. Such increases most certainly influence the ability of even moderate flows to entrain and suspend sediments. In fact, Gordon (1975) measured naturally occurring bursts in a tidal estuary and found that the burst contribution to bottom stress, turbulence production and sediment transport was significant. Apparently, the effect of bursting was manifest as a 4- to 5 fold increase in Reynolds stress, resulting in intermittentsediment transport by traction and suspension.

BURSTING AND THE ORIGIN OF BAHAMIAN WHITINGS

Demonstration of the significance of burst cycles in turbulence production (Kline *et al.* 1967; Kim *et al.* 1971;Grass 1971) and the intermittent transport of sediment (Heathershaw 1971; Gordon & Dohne 1973; Gordon 1975) is an important contribution toward understanding the whiting phenomenon in the Bahamas. Whereas much of the knowledge of fluid boundary-layer structure and dynamics

is derived from experimental flume studies, integration of these ideas with available chemical, physical and geological data for whitings suggests that whitings may resonably be interpreted as resuspended fine-grained bottom-sediments. A number of the observed physical characteristics of whitings are consistent with predicted and observed pat-

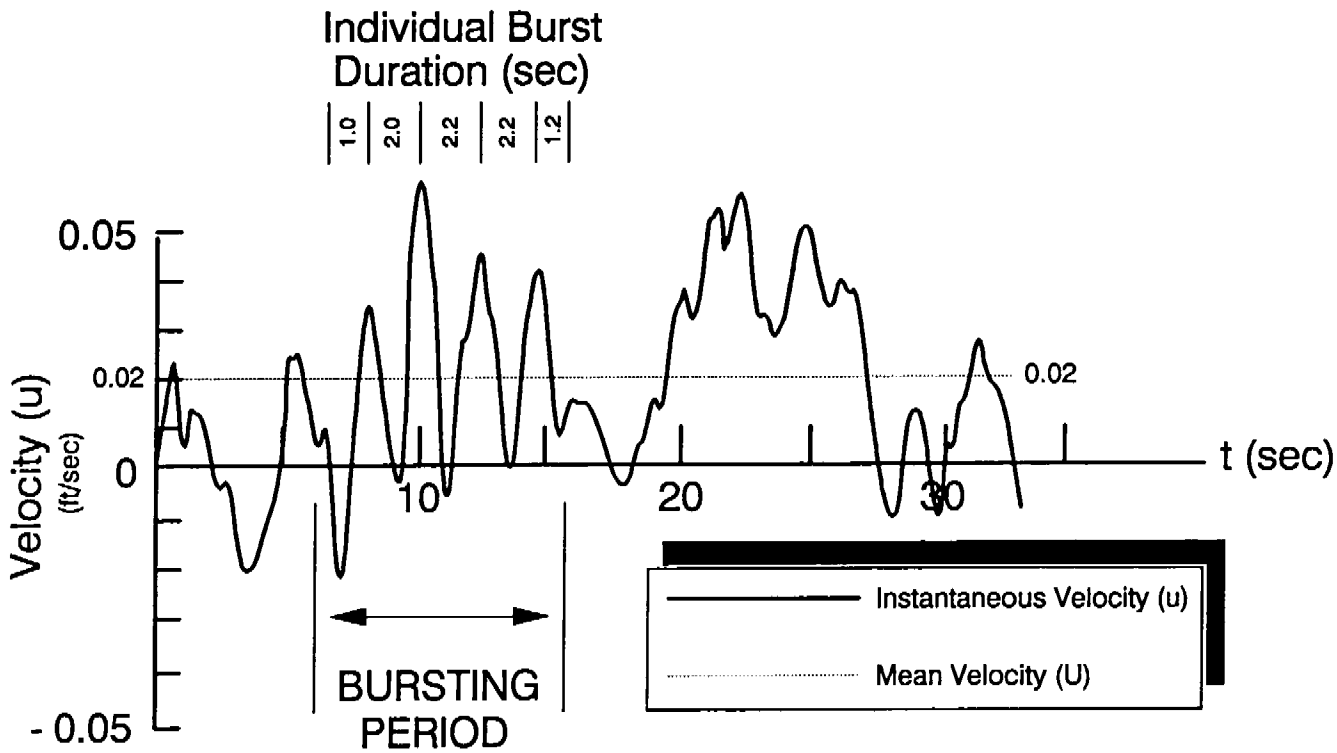


Figure 8. Time-series of velocity fluctuations on which bursting phenomena can be identified. Maximum velocity excursions in this example are about three times the mean velocity for this relatively low velocity flow. Reported excursions up to 4 times mean velocity are known for faster moving fluids (U = 14.5 cm s $^{-1}$; Grass 1971). In addition, notice that the bursting period has duration of about 8 10 seconds during which several pulses of high velocity fluid may be observed. Numbers above observed burst period represent-duration of individual bursts in seconds. Note that individual bursts are not only frequent, but also of large magnitude. Thus, the bursting process is a characteristic feature of turbulent flow systems (from Kim et al. 1971).

terns of turbulent-flow systems. For example, the frequent occurrence of subparallel, digitate zones of turbidity within whitings (Shinn *et al.* 1989) is reminiscent of the characteristic alternations of high- and low-velocities observed in cross-stream profiles (Fig. 6) which resulted in alternating regions of sediment transport and redeposition in flume experiments (Kline *et al.* 1971). Also, the visible, roiling nature of whiting water parcels (Cloud 1962a; Shinn *et al.* 1989) may be the expression of the bursting process as water moves across Great Bahama Bank during the tidal cycle.

Mean tide-dominated flow velocities (U) over Great Bahama Bank in the region of abundant whitings are reported to range from 8 - 50 cm s^{-1} (Smith 1940; Cloud 1962a; Shinn *et al.* 1989), similar to velocities used for laboratory experiments of turbulent flow over smooth and rough boundaries (Grass 1971). For this range of velocities, peak flow rates during bursting ($\simeq$ 4U) may be expected to fall in the range 32 - 200 cm s $^{-1}$. Neumann *et al.* (1970) demonstrated threshold sediment-transport velocities below 25 cm s^{-1} for natural carbonate sands from the Bight of Abaco, Bahamas. Thus, expected burst-velocities should be capable of suspending all particles smaller than 0.125mm, provided such size ranges are available in the sediment (Sengupta 1979). This observation becomes particularly important when the geographic distribution of whitings is compared to Bahama Bank sediment facies (Fig. 1; Purdy 1963).

As previously stated, whitings are significantly correlated with pellet-mud and mud sediment facies and notably absent from other regions of the Great Bahama Bank. An examination of Purdy's (1963) original data (Table 1) shows why this is so - the average weight percents of sediment <0.125mm in pellet-mud and mud facies are 42.8% and 61.7% respectively. The weight percent of sediment <0.125mm in remaining bank-marginal facies is generally less than 5%. Thus it would appear that whiting frequency is related not only to available environmental energy, but also to the availability of fine-grained sediment (additional support for a resuspension origin of whitings comes from the northern shelf lagoon of Belize where whitings grade southward into "brownings", a color change believed to reflect increasing quantities of suspended terrigenous mud associated with the Belize Delta; E.G. Purdy, personal communication, 1990). Purdy (1963, p.495) suggested that sediment facies distributions on Great Bahama Bank approximated an idealized gradient from high-energy sands near the bank margin to low energy muds on the bank center. The geographic distribution of whitings on Great Bahama Bank can also be related to this concept of "enviromental energy

facies" (Fig. 9). Bank marginal depositional environments are certainly sufficiently turbulent to resuspend bottom sediments (i.e. tidal-current and wind-wave energy is high), yet whitings are not often observed in these settings because bottom sediments are virtually devoid of fine-grained carbonate. Whitings are apparently infrequent as well in a bank-top area close to the western shore of Andros Island. Here, carbonate mud is abundant but sufficient energy for sediment resuspension is generally lacking (i.e. environmental energy flux is low) because this region lies in the leeward "energy shadow" created by the island. Only on the central bank-top is energy flux sufficiently high and carbonate mud sufficiently abundant for whitings to originate.

BIOLOGICAL INFLUENCES ON BOTTOM-SEDIMENT STABILITY

The availability of mud over the broad areas covered by pellet-mud and mud facies coupled with environmental energy gives rise to some astonishment that the entire bank-top is not continuously "whited" with suspended sediment. The explanation for the absence of whole-bank whiting appears to lie in the known bottom-stabilizing influence of subtidal microbial mats (Bathurst 1967; Neumann et al. 1970).

These mats, formed of communities of algae and cyanobacteria, were investigated by Neumann et al. (1970). Results of underwater flume experiments indicated that minimum velocities necessary to induce sediment erosion and suspension were significantly larger (2 to 5 times) when bottom sediments were inhabited by these communities. Thus, in the presence of bottom-stabilizing microbial sheets, bottom sediments may resist entrainment even for instantaneous "bursting" velocities of 100 - 200 cm s^{-1}. The apparent widespread occurrence of subtidal microbial sheets in the Bahamas region is thus an important factor to consider when describing sedimentological aspects of whitings. The ubiquitous occurrence of subtidal microbial mats on central Great Bahama Bank (Newell *et al.* 1959; Purdy 1963; Bathurst 1967 & 1971) may relate to restriction of browsing, grazing and bioturbating organisms within the region of elevated salinity. It is likely that extensive mat development is an impediment to formation of bank-wide whitings, except during periodic high-energy events (storms) which may disrupt mats over a large area and resuspend large quantities of bottom sediment.

In addition, it is important to note that whereas subtidal microbial mats are ubiquitous elements of the bank top ecosystem, these mats are not continuous. The occurrenceof individual whitings suggests local disruption of subtidal mats with subsequent bottom sediment destabilization and resuspension during turbulent flow. A variety of factors may contribute to destruction of mats, including intermittent high flow velocities and the activity of schools of bottom-feeding fish or other bioturbators (e.g. *Callianassa sp.* whose penetrative burrow mounds may disrupt mat continuity) and normal, cyclic mortality of mat-building communities. Thus, the effectiveness of sediment stabilization by microbes can be viewed as a balance between those factors which contribute to microbial mat development (e.g. elevated salinity and consequent restriction of browsing organisms) and microbial mat destruction (e.g. feeding activity of fish, bioturbation, periodic storms, mat life cycles). While it is not precisely known how these factors interact or contribute to the formation of whitings on Great Bahama Bank, these factors must be considered in any analysis of the origin of whitings.

CONCLUSIONS

The origin of whitings has perplexed generations of scientists conducting research on sedimentary carbonate environments. However, an examination of the distribution of whitings on northern Great Bahama Bank indicates a non-random pattern of occurrence related to the geographic limits of pellet-mud and mud sediment facies (Fig. 1). Thus, whiting formation is controlled in part by the availability of nonpelleted, fine-grained sediment (mud contents range from 43% to 62% by weight in muddy, bank-interior facies but are <5% by weight in bank-marginal facies; Table 1).

Available chemical data are interpreted to indicate indirect support for resuspension of bottom sediment as the origin of whitings. Bulk chemistry (pH, total alkalinity, relative aragonite saturation) of waters within the area of frequent whitings showed that large-scale instantaneous precipitation was unlikely. In addition, recall that ^{14}C activity of fine-grained sediment collected in whitings yields values indicative of formation prior to suspension in the whiting. The most likely source of such "old" calcium carbonate on the central bank-top is bottom sediment. Indeed, ^{14}C age determinations for bottom sediments are in accord with ^{14}C-ages determined for fine-grained carbon
ate filtered from whiting water parcels.

Recent observations of fine-grained carbonate associated with degraded organic matter are relevant to issues surrounding the genesis of lime mud. However, the precise relation of these observations with respect to whiting origins is tenuous; it is not clear how organic matter/carbonate associations relate to known temporal, geographic, physical and sedimentological characteristics of whitings.

Bottom sediment resuspension is an unappreciated though potentially significant component in whiting generation in the Bahamas. An evaluation of the dynamics of boundary-layer turbulent flow indicates that turbulence production is generated primarily by transient bursts during which instantaneous velocities may exceed mean flow by

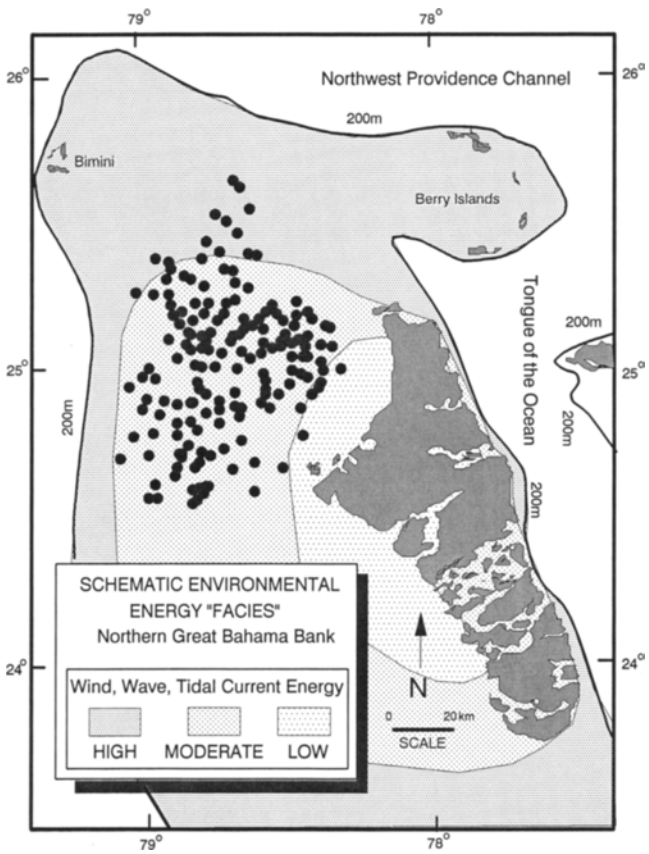


Figure 9 Schematic figure of environmental energy "facies" of Great Bahama Bank (after Purdy 1963). Environmental energy is defined as the sum of tidal current, wind and wave energy. Energy near bank margins is high but decreases progressively across the bank interior toward the leeward coast of Andros Island. Note that whitings (closed circles) are uncommon near bank margins where energy is high. Although bank-marginal environments are sufficiently turbulent for whiting formation, bottom sediments in bank-marginal facies are nearly devoid of fine-grained carbonate. Near Andros Island, carbonate mud is abundant in bottom sediments, but whitings are rare owing to creation of an "energy shadow" by the island. Whitings are only common on the central bank interior where energy is sufficiently high and carbonate mud sufficiently plentiful for resuspension to occur.

as much as 4 times. For tidal flow on the Bahamas bank-top, this means that peak velocities in the range 32 - 200 cm s^{-1} are possible and would be accompanied by large-magnitude variations in shear stress and lift force over rough bottoms. It must be concluded from these data that resuspension of fine-grained (<8 um) bottom sediment similar in composition to that found suspended in whitings is a reasonable consequence of boundary layer bursting processes. Physical features of individual whitings (e.g. digitate leading boundaries, roiling fronts) are similar in many respects to features observed within turbulent-flow systems. Great Bahama Bank may be subdivided into environmental energy "facies", wherein energy is defined in qualitative terms as the flux of tidal-current, wind and wave energy. It can be seen (Fig. 9) that whitings are persistent features only in those areas of the bank-top where energy flux is sufficiently high and carbonate-mud sufficiently abundant for their generation.

Finally, the bottom-stabilizing influence of subtidal algal/cyanobacterial mats was considered. Previous studies showed that minimum flow velocities required for erosion and suspension of bottom sediments were significantly greater when mats were present. It is postulated that ubiquitous mat development stabilizes sediments in the pellet-mud and mud facies (preventing bank-wide resuspension events), but that discontinuities where mats are locally disrupted permit turbulent flow processes to interact directly with bottom sediments, creating small-scale (areally restricted) whitings. That is, the destruction of mats by various processes exposes fine-grained bottom sediments to the bottom destabilizing action of turbulent tidal flow on the bank-top. Processes which may contribute to algal-mat degradation include periodic high-energy episodes (e.g. storms or high-wind events), feeding activity of bottom-feeding fish, grazing organisms or other bioturbators and natural, cyclic aspects of mat mortality.

While the frequency and efficiency of many of the above processes is not presently known, the important role of physical processes seems to beg recognition. Indeed, in view of available chemical and biological data, bottom sediment resuspension appears the most parsimonious explanation for whiting phenomena; an explanation which does not require extraordinary chemical or biological events or microenviroments. However, it is clear that a complete understanding of Bahamian whitings will require an integrated, interdisciplinary view of the various physical, chemical, sedimentological and biological components of the Great Bahama Bank environment as well as rigorous comparison with other whiting phenomena worldwide.

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